

Eduardo Hernández Valdez

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EDUCATION

ITESM Av. Eugenio Garza Sada 2501 Sur, Tecnológico, 64700 Monterrey, N.L. Mexico
Engineering in Robotics and Digital Systems; Avg 92/100 - GPA 3.6 Aug 2022 - June 2026

RESEARCH EXPERIENCE

Joint Robotics Laboratory (JRL) @ CNRS-AIST Tsukuba, Japan
Research Intern - Humanoid Robotics - In preparation for RA-L (submission planned Jan 2026) Aug 2025 - Feb 2026

- Developed **GPU-accelerated** tactile sensors in **JAX** for robot simulation, achieving real-time performance
- **Tactile-guided reinforcement learning** policy development for **dexterous manipulation**
- **Hardware & software co-design** of tactile matrixes for dexterous humanoid **robot teleoperation**
- Microcontroller interfacing **PCB design** for reading 300+ analog signals and transmitting them over UDP

FLOWMAS: Multimodal Flow Matching for Trajectory Generation Mty, Mexico
Primary Author - Independent Research @ ITESM - Manuscript in preparation Sep 2025

- Designed **multimodal CNF model** with **spatial memory** for RGBD-conditioned trajectory synthesis

Real-time in-situ defect detection for fresh produce using Artificial Intelligence Mty, Mexico
Team Researcher Feb 2024 - June 2024

- Developed CNN architectures to improve accuracy and speed of already existing algorithms using **ONNX**

LEADERSHIP

Vice-President @ VantTec Robotics Club Mty, Mexico
Managed technical direction and R&D roadmaps of 80+ team members on 5 different projects Aug 2025 - June 2026

Teaching Coordinator - Robotics/AI Classes Mty, Mexico
Designed and delivered machine learning curriculum for high school students Feb 2025 - June 2025

Self-Driving Vehicle Project Leader @ VantTec Robotics Club Mty, Mexico
*First author on **accepted ICRA 2025 Poster*** Aug 2024 - June 2025

- End-to-end local planner of autonomous vehicles using **generative flow models**
- Designed **custom PCB** and embedded firmware for **autonomous vehicle** actuation system

Unmanned Surface Vehicle R&D Team Leader @ VantTec Robotics Club Mty, Mexico
*Roboat 2024 & 2025 **Finalist team*** Aug 2022 - Feb 2025

- On-device object detection inference framework using **TensorRT** with **stereo triangulation** for 3D localization
- Configured **Linux systems, Docker, ROS2 and hardware drivers** for our vehicle's infrastructure

WORK EXPERIENCE

Computer Vision Intern - AMI Automation May 2025 - Nov 2025 (Remote Aug-Nov)
• Integration, research and **SIMD** optimization of computer vision algorithms for the steel **industry**

Freelance General Purpose Programming - fiverr.com/eduarch42 Summer 2019, 2020, 2021, 2022
• Developed solutions in **Data Science**, Backend Development, GUI Applications, with an overall 4.9/5 rating

PROJECTS

Tensorgba - CNN interfacing with Gameboy Advance games. (Self-driving Mario Kart)

Embedded EKF for Pose Estimation - Drone interfacing MCU running **estimating filters on sensor readings**

Waypoint Car - Hobby RC car estimating it's position with **embedded sensors** and **custom electronics**

GPU accelerated camera stitching - Camera extrinsic and intrinsic **calibration** for lab showoff

AWARDS & ACHIEVEMENTS

Xignux Challenge 2024 - Top-3 National - Technology **entrepreneurship proposal** with a focus on sustainability

HackMTY 2022 Hackathon - 1st Place - Hardware box designed to store online identity & cryptocurrencies

International Collegiate Programming Contest – Top 7 within ITESM MTY.

International Science Competition @ ITESM – Finalist in Computer Science category.

SKILLS

Programming: *Proficient:* Python, C, C++, Go. *Intermediate:* JS, Matlab, Rust. *Beginner:* CUDA, Verilog

Technologies/Software: Linux, ROS[1,2], PyTorch, JAX, Tensorflow, TF-Lite, Numpy, OpenCV

Languages: Spanish (Native), English (TOEFL - 103/120), German (Elementary), Japanese (Learning)